MS2 Summary: FlowGuard

1 What I achieved in this milestone

- **What works now:** an hourly GitHub Actions pipeline. The job `ingest` pulls BAFU and Open-Meteo data into a Cloud Storage bucket (append-only Parquet, release state and fetch time kept per row). A second job, `features`, then builds the feature table (156 columns per station, one file per station and month, plus a manifest). Backfill: about 1.45 million BAFU rows since 2011 and 3,244 weather runs since 07.08.2024. A resumable backfill command, a Docker image, CI with ruff and 299 tests (no network in tests) are in place.
- **Scope change after the data probe:** I forecast three stations (Andermatt 2087, Schächen 2491, Seedorf 2056), not six. 2492 and 2649 are diversion flows with discharge only, so they are inputs. 2299 (Alpbach) is not live, so it is only collected. 2049 was the wrong station. The river network stays: Andermatt and the Schächen feed Seedorf.
- **Target and baseline:** the 24 h change of the water level in cm, added to the current level. The persistence baseline (tomorrow equals today) has a validation/test MAE of 4.04/3.02 cm at Andermatt, 2.70/1.87 at Bürglen and 9.77/11.96 at Seedorf (`docs/eda_notes.md`). Because of this I changed the goal: better than persistence at every station, 3 cm only at Andermatt, at most 80 percent of persistence at Seedorf.
- **Weather:** forecasts as issued (ECMWF IFS runs), taken only from runs that existed at the time. 11 runs do not exist at Open-Meteo, so the pipeline falls back to the newest earlier run or leaves the value missing. Weather features start on 10.08.2024, so weather training has about 2.2 years, not the 15 years of level data.
- **No look-ahead:** the cleaning checks have a causal mode for the features, and tests check that cutting the data at time t or changing the data after t never changes the features at t.
- **Data trust:** BAFU validates data only months later. I keep release state and fetch time for every row and report the unvalidated part of the test period separately. Train is 1 percent unvalidated, validation 60 to 100, test 100.
- **Look at first:** README, `docs/features.md`, `docs/eda_notes.md`, `src/flowguard/common/`. Actions: ingest.yml, 26 scheduled runs since 28.10. The six since 02.11 (#23 to #28) run both jobs, ingest and features, and are all green.
- **What it cost:** half of the planned stations, a weather window of 2.2 years, three days of weather features at the start, and about a week for probes and the storage design before any model. GitHub starts the hourly job less often than hourly (about 4 runs a day instead of 24), which the 48 h refetch covers. The validation and test periods are quiet, so 8 of 18 risk bands are thin and reported as not evaluable. There is no model yet; that is MS3.

2 Response to MS1 reviews

Every label has an entry. Commit hashes are from `git log` of 03.11.

Label	Status	My answer	Commit or file
E1	Changed	The README now names the four ID2223 projects and says what FlowGuard does differently: hourly 24 h horizon, a connected gauge network with diversion flows, joint scoring of the high and low band against persistence, weather as issued, documented split and release state. I do not claim trend features or the chronological split as new.	README, related work (`f5e385f`)
A1, B1	Partly	Agreed. Open-Meteo precipitation and temperature are in the feature table: observed over the last 24 to 72 h and the forecast for the next 24 h, from runs that existed at the time (37 weather columns). The comparison with and without weather, including the largest 24 h rises, is reported with the model in MS3.	`e7bff10`, `360d8ce`, `491b486`, `27951d8`, `docs/features.md`
E2	Changed	Confirmed. On 01.10 the repo had only a one-line README. It now explains the project, the data, the pipelines, how to run it and the known limits.	`0b3c474`, `f5e385f`

Label	Status	My answer	Commit or file
A2, B2	Changed	Several upstream inputs per station are in the config (up_main, up_trib, div_q). I measured the travel time by cross-correlation: the best lag is 1 to 2 h and the link is gone after about 6 h (r 0.52 at 2 h for 6 h changes). The planned 1, 6 and 24 h differences cover this, so I added no extra lag. The two diversion gauges are unrelated to each other and weakly related to Seedorf, so I keep them but check their importance in MS3.	`9e94895`, `7e593c5`, `67ceab9`
A3	Changed	Agreed. Three chronological segments with the same cut dates for all stations, set in config, and an embargo between the segments. A row belongs to a segment only if t and t+24 h are inside it. Validation selects the model, the test is scored once.	`af7f576`, `config/pipeline.yaml`
A4, B3	Partly	Agreed. Persistence MAE per station, segment and risk band is in `docs/eda_notes.md`. It showed that 3 cm is already met by persistence at Bürglen, so the goal is now: better than persistence at every station, 3 cm at Andermatt, at most 80 percent of persistence at Seedorf. The model MAE as a ratio to persistence follows in MS3.	`7e593c5`, `docs/eda_notes.md`
B4	Partly	Agreed. The risk-band percentiles come from the training segment only; the baseline analysis already uses this rule. The percentiles are stored with the model in MS3.	`7e593c5`
E3	Changed	BAFU delivers the hourly series per station and parameter, up to 10,000 rows per query, which I handle with 30-day windows. The backfill holds about 1.45 million rows since 2011 (1,158 requests). Fallbacks are BAFU's monthly bulk files, hydrodaten.admin.ch and existenz.ch. They are named but not tested. The 11 missing weather runs are listed.	`d4e33e3`, `491b486`, `docs/data_notes.md`
E4	Partly	GitHub Actions gives me the schedule and the tests next to the code. Docker gives the same environment locally, in CI and on Cloud Run. MLflow will run on DagsHub and is set up in MS3; until then it is a local file store.	README, why this stack (`f5e385f`), `2271afa`
E5	Changed	KW Schächen (2649) and the second diversion (2492) have discharge only and follow plant operation, so they are inputs and never forecast targets. This keeps them out of the pooled model's error.	`config/stations.yaml` (`9e94895`)